

# Shubham Kumar

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## EDUCATION

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Indian Institute of Science, Bangalore  
*Master in Artificial Intelligence*

Bangalore, India  
*August 2019 - June 2021*

Indian Institute of Technology, Kanpur  
*Bachelor in Electrical Engineering*

Kanpur, India  
*August 2012 - April 2016*

## EXPERIENCE

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IPsoft Global Services  
*RGD Engineer · IPsoftdigital*

Bangalore, India  
*Jan 2023 - Current*

- **Generative AI:** Worked on two application based on Generative-AI using openAI APIs, LangChain and ElasticSearch.
  - \* EazyML-Copilot Assistant that could answer questions related to EazyML APIs. It's been accomplished using creating data based on EazyML APIs and Fine-tuning open-AI model on top of that.
  - \* Question-Answering System for PDF One could search through multiple pdf-documents and it provides exact answer for asked question and it also highlight section in pdf from where answer has been found with page number and paragraph.
- **J&J :** Worked on three different projects for JGJ.
  - \* Point Distance Correction of ZOI Worked on APIs implementation for XAI and Unified Confidence Score module. Experimented with many different approaches for confidence score model and point distance, considering input-output features and model architectures.
  - \* Strategic Priority Prediction Created Rules and ML based model to prediction of Strategic Priority. Used different Generative AI models(from OpenAI and Google) to predict Strategic Priority, Exit Timing and reason for these predictions. Also, used multi-agent system from framework such as Langchain and DSpy to improved accuracy.
  - \* GDInsight Contributed to time and memory performance optimizations, resulting in significantly faster API responses. Collaborated with a team of 89 backend developers to implement features such as *dynamic SQL query generation* and search result retrieval, leveraging LangGraph along with AWS Bedrock and Azure OpenAI APIs.
- **EazyML PIP packages:** Deployed various packages for EazyML such eazymml, eazymml-xai, eazymml-cf and eazymml-dq and eazymml-genai on pip, *documented* and *published* it on *ReadTheDocs*. Implemented the integration of package APIs using license keys and developed a backend system to track API usage statistics by users.
- **Audio-Avatar Video Chat Application:** Our Audio-Avatar Video Chat Application is a versatile and user-friendly platform designed to facilitate seamless communication through for personal or professional use. Option to select many different Video Avatar model based on LipGAN, Wav2Lip, SADTalker, RAD-Nerf, GeneFace and R2-Nerf in order to have better speed or Video Quality.
- **Unsupervised clustering:** Implemented custom unsupervised clustering method based on cluster density and Hierarchical clustering based algorithm. It takes care of number of optimal clusters, relevant features and optimal density for each cluster. We could pre-define some hyperparameters like number of iterations, minimum cluster density or samples.

Quantiphi Analytics Pvt Ltd  
*Machine Learning Engineer RGD · RGD India*

Bangalore, India  
*July 2021 - Oct 2022*

- **Knowledge Graph Question Generation:** Generating question-answer pairs using knowledge graph(KG) triples as input. Fine-tuned GPT2 Decoder model from hugging-face and taking input as KG triples, semantic and triples word embedding from HF embedding Layer. Trained TransE model to generate entity and relation embeddings over CORD-19 dataset
- **Disease Diagnosis:** Disease Diagnosis refers to the process of decision-making which will lead to an accurate understanding of a patient's health problem. Implemented Neural Network and SVM model which takes one-hot, dense embedding or node-embedding to predict probable diseases from given symptoms name or symptoms definitions.
- **Deep Search:** Implemented Paragraph Retrieval and Knowledge graph search for retrieving the relevant documents using ElasticSearch. Paragraph Retrieval Search is a combination of phrase, bi-gram, keyword search with reranking using the cross encoder. In KG Search, Searched over the knowledge graph triplets, including metadata and retrieve the relevant triplets and used filters to refine the search. Published a paper in IEEE Big Data 2022

- Responsible CV: Implemented Non Invasive Demographic Detection, a novel way of making use of the remaining part of the human body and still be able to predict gender and age of people. Implemented various explanation method such as GradCAM, SHAP, IG, XRAI, for classification model and IG, XRAI for object detection model

DoubleSpring Media Pvt Ltd  
Mobile Application Developer · Teamwave

Bangalore, India  
Feb 2017 - June 2018

- Teamwave: Developed Customer relationship management module for Android and iOS Operating System from scratch and guided other Android Developer in developing Project Management Module for Android platform

GlobalShiksha Pvt Ltd  
Android Developer · Educhat

Bangalore, India  
Feb 2017 - June 2018

- Educhat: Worked on designing the UI for the mobile platform. Maintained and improved the performance of the existing system by storing data and updating it so we could can run it offline

Foodmonk  
Co-Founder and Mobile Application Developer · Foodmonk

Kanpur, India  
May 2015 - June 2016

- Foodmonk: Worked in Designing, Coding and Testing Phases of mobile front-hand and back-hand portion by using Json parsing, AsyncTask, DB Helper and Service. Achieved a daily average of 30 orders within first 15 days

## PROJECTS

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Large Scale Complex Information Retrieval on Knowledge Synthesized from CORD-19: Published paper on IEEE Big Data 2023. We extract helpful information from unstructured text data and perform complex information retrieval using three major components- Paragraph Retrieval, Triplet Retrieval from Knowledge Graphs, and Complex Question Answering (QA).

GCP ACE Certification: Passed Google Cloud Associate Cloud Engineer certification Exam. Learned skills of deploying, monitoring, and maintaining projects on Google Cloud.

Video Magnification using Neuromorphic Camera: Worked on video magnification using neuromorphic Camera based on minute brightness or motion change.

Selfie to Anime image translation: Implemented an unsupervised image-to-image translation using Least Squares GAN and AdaLIN Normalization function.

## PROGRAMMING SKILLS

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Languages: Python, C++, JavaScript, Java, Swift    Technologies: Tensorflow, Pytorch, Machine Learning, GCP